Elimination 1: Visual Fidelity Score (VFS)

Judging frames

To check whether the last frame (in case of Progressive Reveal, Colour Pop, Hopping Bounding Box, Sliding Bounding Box) and all the frames in case of Alpha Masking are faithful to the original image or not

Judge: Qwen 235B -A22B

Have general rules and attune to style-specific adapters

Implementation Idea: The generic instructions at the start of the prompt and before and after the style adapter are common in every prompt. The text of the style adapter is added based on the chosen style for the animation. So the gating must be done accurately based on style

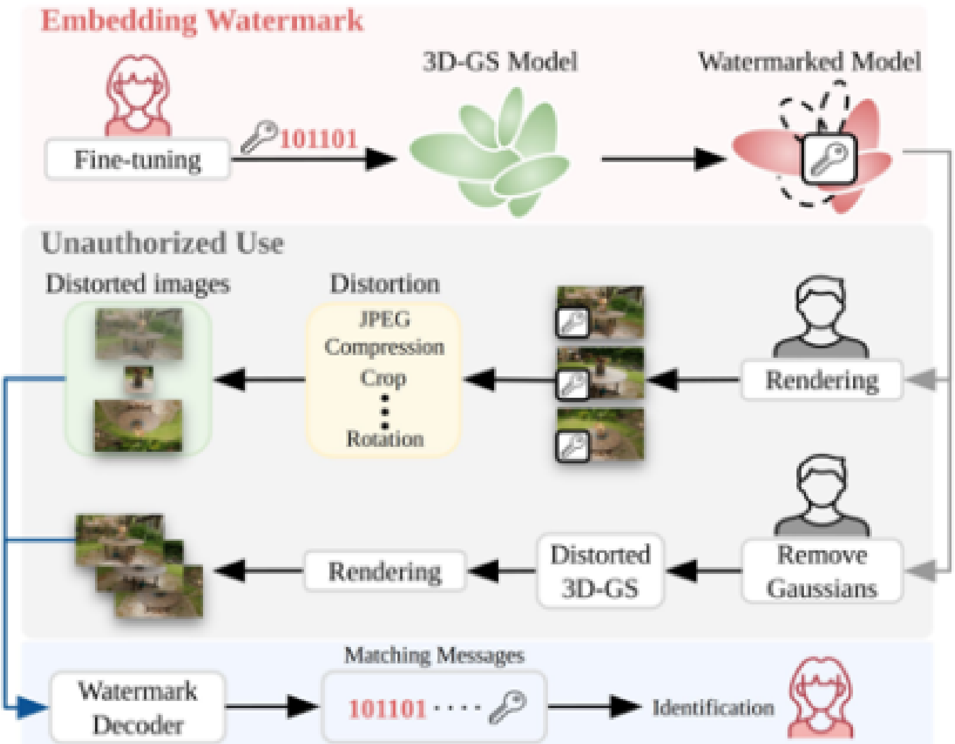
Progressive Reveal

```
You are an expert Diagram Animation Evaluator. Your task is t
o compare the provided animation frame(s) against the origina
l source diagram to assess its visual fidelity.

You MUST evaluate the frame(s) strictly according to the foll
owing 6 criteria:

1. Element Alteration: Are all corresponding elements from th
e original diagram present, undistorted, unmorphed, and unalt
ered in the given animation frame/frames? Any style-specific
visual quirks (e.g., masking opacities, highlight boxes) MUST
be tolerated and not penalized as distortion or morphing.
```

```
2. Visual Layout Fidelity (CRITICAL): Is the overall visual layout in terms of containment boxes, element placement, and directional flows captured exactly as in the source diagram?
3. Geometric Layout Fidelity (CRITICAL): Do elements preserve their correct relative positions, spacing, and alignment?
4. Style & Color Fidelity: Do the colors, aesthetics, and visual styles match the source diagram (accounting for the animation style)?
5. Proportion & Scale Fidelity: Are element sizes, aspect ratios, and relative scales consistent?
6. Text & Typographic Integrity: Are text labels and mathematical symbols perfectly preserved without corruption, truncation, or hallucination?

Be extremely strict in your evaluation. Any structural mismatch, missing arrow, misplaced containment box, or corrupted text label must lead to a noticeable drop in score.
The CRITICAL factors must induce a heavier penalty on the score.
The following is NOT a penalty and MUST NOT lower any score:
  - Additional text panels, equations, step-number overlays, titles, decorative
    annotations, or instructional labels added by the animation system. These are
    considered additive helpful content.

*** STYLE ADAPTER: Progressive Reveal ***
In Progressive Reveal, elements of the diagram are cumulatively added to the canvas one group or one element at a time. Once revealed, elements remain visible for the rest of the animation. Nothing is hidden or masked after being shown.
You are provided with the original diagram followed by the LAST frame of the animation sequence (which represents the completed diagram). Evaluate whether this fully revealed end result is completely faithful to the original source diagram.
```

````
*** OUTPUT REQUIREMENT ***
Output ONLY valid JSON matching the schema below. Reason deeply through each factor in the `assessment` dictionary first, then assign a single `visual_fidelity_score` on a scale of 0.0 to 10.0 for each frame evaluated.

```json
{
  "frame_evaluations": [
    {
      "frame_name": "string (filename of the frame evaluated)",
      "assessment": {
        "element_alteration": "Detailed reasoning regarding element presence, distortion, or style tolerance.",
        "visual_layout_fidelity": "Detailed reasoning regarding containment, flow, and structural layout.",
        "geometric_layout_fidelity": "Detailed reasoning regarding relative positions, alignment, and spacing.",
        "style_color_fidelity": "Detailed reasoning regarding colors, line weights, and visual matching.",
        "proportion_scale_fidelity": "Detailed reasoning regarding element sizes and aspect ratios.",
        "text_typographic_integrity": "Detailed reasoning regarding text legibility, corruption, or hallucination."
      },
      "visual_fidelity_score": 0.0
    }
  ]
}
```

Do not output any text before or after the JSON block.
````

Original Image



Original image

Gemini 3.1 Pro generated animation

attachment:a0487e24-1b67-4596-a1c1-7037c6d6a7dd:CVPR_2025_pipe00010.mp4

Gemini 3.1 Pro generated animation


```
{
  "frame_evaluations": [
    {
      "frame_name": "animation_frame_2.png",
      "assessment": {
        "element_alteration": "Critical missing element: The 'Distortion' box (containing JPEG/Compression/Crop/Rotation list) is completely absent in the animation frame. The 3D-GS Model icon is morphed from leafy shape to bubble clus
```

```
ter, and Watermarked Model
icon is altered from flowe
r/magnifying glass to loc
k/ribbon. These are struct
ural changes, not style qu
irks.",
        "visual_layout_fid
elity": "Structural mismat
ch: Original has 3-step di
stortion process (Distorte
d images → Distortion box
→ Distorted images), but a
nimation frame skips the D
istortion box entirely. Fl
ow direction is broken: Di
storted images directly co
nnects to Rendering. Conta
inment boxes for sections
are misaligned with origin
al structure.",
        "geometric_layout_
fidelity": "Relative posit
ions severely distorted: W
ithout the Distortion box,
the middle section is comp
ressed. The 'Distorted ima
ges' element is misaligned
with Rendering step (shoul
d have intermediate elemen
t). Spacing between elemen
ts is inconsistent with or
iginal proportional relati
onships.",
        "style_color_fidel
ity": "Iconography mismatc
h: 3D-GS Model changes fro
```

```
m organic leaf shape to ab
stract bubbles. Watermarke
d Model changes from diagn
ostic tool to security bad
ge. Color scheme of sectio
n backgrounds (pink vs ori
ginal light red) is incons
istent. Line weights and a
rrow styles differ in crit
ical flow paths.",
        "proportion_scale_
fidelity": "Aspect ratios
altered: 3D-GS Model's bub
ble cluster has different
width/height ratio than or
iginal leaf shape. Distort
ed images container is sca
led differently without in
termediate distortion ste
p. Relative sizing of Wate
rmark Decoder element is i
nconsistent with origina
l.",
        "text_typographic_
integrity": "Critical text
omission: Entire 'Distorti
on' box text (JPEG/Compres
sion/Crop/Rotation) is mis
sing. Binary string '10110
1' appears correct but is
positioned differently rel
ative to other elements. S
ection headers ('Unauthori
zed Use') maintain text bu
t lose contextual alignmen
t."
```

```
      },
      "visual_fidelity_score": 2.0
    }
  ]
}
```


AnimateBanana SVG video

https://drive.google.com/file/d/1UgU7CyobEEGNlas1gdZlpE1jdloyKVJn/view?usp=sharing


```
{
  "frame_evaluations": [
    {
      "frame_name": "frame_2.png",
      "assessment": {
        "element_alteration": "Multiple critical element alterations: (1) The embedding key icon is changed from a simple key to a keyhole design, (2) The female creator icon is simplified from detailed illustration to minimalist head silhouette, (3) The Distortion box text is corrupted with added '...' between 'Crop' and 'Rotation' (original had exactly four items without ellipsis), (4) The Watermarked Model replaces the original key icon with a key badge inside the model, (5) Matching Messages adds a key icon next to the bit string. These are structural changes beyo
```

```
nd style quirks. The botto
m caption is additive cont
ent and not penalized.",
        "visual_layout_fid
elity": "Critical structur
al mismatch: The 'Distorte
d images' label is incorre
ctly positioned only to th
e left of the top image st
ack instead of centered ab
ove both stacks as in the
source. The directional fl
ows (embedding → unauthori
zed use → verification) an
d containment band structu
re are preserved, but the
misaligned label creates a
n incorrect information hi
erarchy. Colored backgroun
d bands (pink/gray/blue) a
re tolerated as additive s
tyle.",
        "geometric_layout_
fidelity": "Significant mi
salignment: The 'Distorted
images' label placement br
eaks vertical alignment wi
th the image stacks. The w
atermarked images maintain
horizontal arrangement, bu
t the top distorted stac
k's label is offset while
the bottom stack lacks lab
eling entirely. Relative p
ositioning of core compone
nts (3D-GS model, Watermar
```

```
k Decoder) is preserved, b
ut the critical label misp
lacement violates geometri
c integrity.",
        "style_color_fidel
ity": "Core color scheme p
reserved (red watermarked
model, green 3D-GS), but p
roblematic alterations:
(1) '101101' bit string ch
anged to red bold text in
embedding step, (2) Key ic
on color altered from blac
k to red outline, (3) Dist
ortion box uses yellow hig
hlight (tolerated as style
quirk). The minimalist ico
n style diverges from the
source's detailed illustra
tions, crossing into eleme
nt alteration.",
        "proportion_scale_
fidelity": "Element scalin
g is generally consistent:
watermarked model and 3D-G
S blobs maintain comparabl
e sizes, image stacks pres
erve relative dimensions.
Minor spacing variations e
xist in the unauthorized u
se section, but no critica
l scale distortions. The v
ertical stacking of waterm
arked images matches sourc
e proportions.",
        "text_typographic_
```

```
integrity": "Critical text corruption: (1) Distortion box adds '...' in operation list (original had no ellipsis), (2) Matching Messages inserts key icon after bit string. All other labels ('Embedding Watermark', 'Fine-tuning', etc.) are preserved verbatim. The '101101' bit string appears in altered styling (red/bold) but content remains intact."
      },
      "visual_fidelity_score": 7.5
    }
  ]
}
```

Original Presentation

attachment:52e726b2-f4ed-4cad-aacb-85604fc31a80:OG_prog.mp4

```
{
  "frame_evaluations": [
    {
      "frame_name": "last_frame.png",
      "assessment": {
        "element_alteration": "All critical elements from the original diagram are present and unaltered in content. The watermark
```

```
key (101101), model icons, distortion operations, and flow components match exactly. Background color variations (light pink top section in animation vs. white in original) and key color (red in matching messages) are tolerated as style quirks per instructions. No distortion, morphing, or missing elements detected.",
        "visual_layout_fidelity": "Containment structure and directional flows are identical. The top 'Embedding Watermark' section and middle 'Unauthorized Use' section maintain the same hierarchical grouping. All arrow connections (e.g., from 3D-GS Model to Watermarked Model, distortion paths to Watermark Decoder) follow the original flow. The blue arrow from Watermark Decoder to Identification is preserved with correct directionality.",
        "geometric_layout_fidelity": "Relative positioning is precise. The 'Distorted images' thumbnails remain left-aligned in both diagrams, with 'Distorti
```

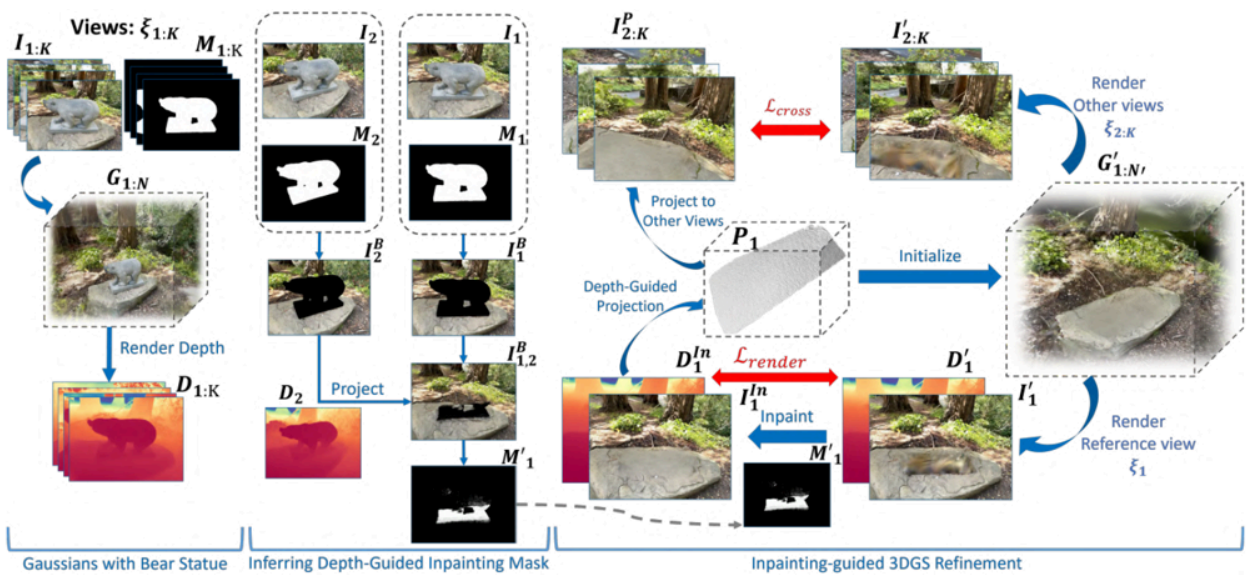
```
on' operations centered ve
rtically between paths. Th
e upper path (Rendering →
Distortion) and lower path
(Remove Gaussians → Distor
ted 3D-GS) maintain consis
tent horizontal spacing an
d vertical alignment. Imag
e thumbnails and icons ret
ain identical spatial rela
tionships.",
        "style_color_fidel
ity": "Core element aesthe
tics match: green 3D-GS Mo
del, pink Watermarked Mode
l, and black arrow styles
are consistent. Tolerated
variations include the top
section's light pink backg
round (vs. white) and red
key text in 'Matching Mess
ages'. These are style qui
rks explicitly exempt from
penalty per instruction
s.",
        "proportion_scale_
fidelity": "Element sizes
and aspect ratios are pres
erved. The 3D-GS Model lea
f, Watermarked Model flowe
r, and image thumbnails ma
intain identical dimension
s. The 'Distortion' box an
d 'Watermark Decoder' comp
onents scale proportionall
y relative to surrounding
```

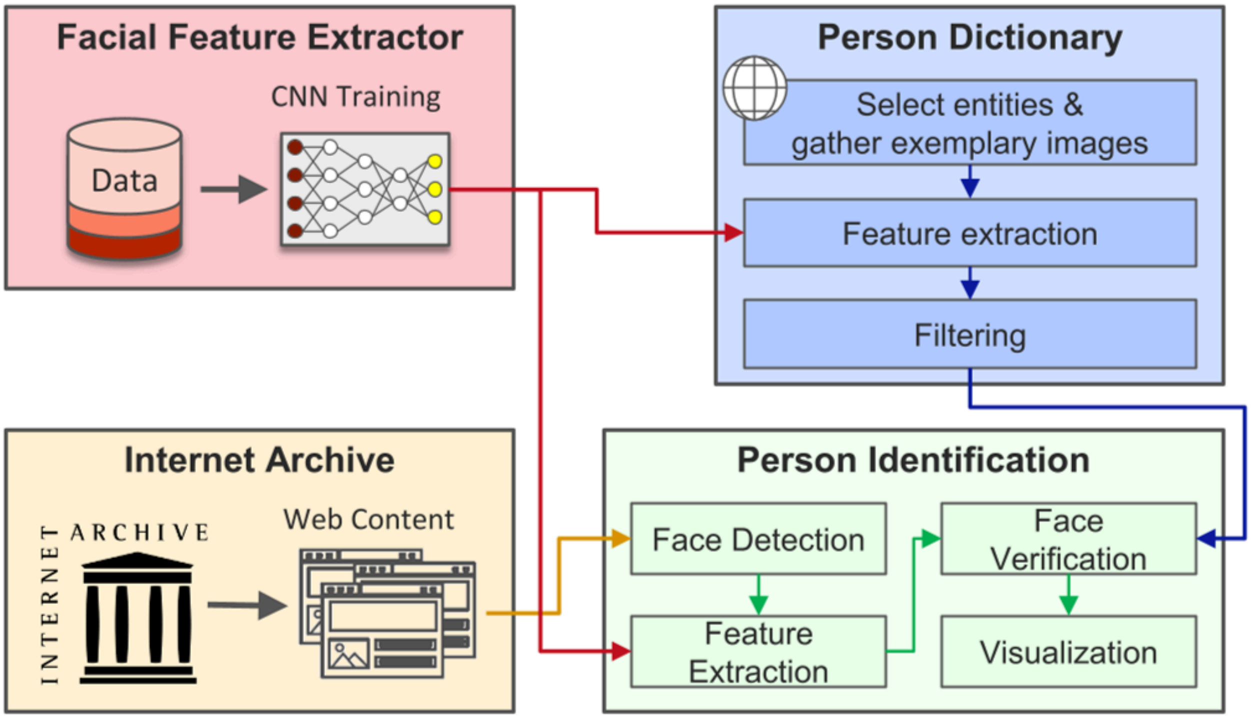
```
elements in both diagram
s.",
        "text_typographic_
integrity": "All text labe
ls are perfectly preserve
d: 'Fine-tuning', 'JPEG Co
mpression Crop ... Rotatio
n', 'Distorted 3D-GS', and
'101101' key sequence matc
h exactly. No truncation,
corruption, or hallucinati
on detected. The red color
for '101101' in matching m
essages is a tolerated sty
le variation."
      },
      "visual_fidelity_sco
re": 10.0
    }
  ]
}
```

Alpha Masking

```
You are an expert Diagram Animation Evaluator. Your task is t
o compare the provided animation frame(s) against the origina
l source diagram to assess its visual fidelity.

You MUST evaluate the frame(s) strictly according to the foll
owing 6 criteria:

1. Element Alteration: Are all corresponding elements from th
e original diagram present, undistorted, unmorphed, and unalt
```

```
ered in the given animation frame/frames? Any style-specific
visual quirks (e.g., masking opacities, highlight boxes) MUST
be tolerated and not penalized as distortion or morphing.
2. Visual Layout Fidelity (CRITICAL): Is the overall visual l
ayout in terms of containment boxes, element placement, and d
irectional flows captured exactly as in the source diagram?
3. Geometric Layout Fidelity (CRITICAL): Do elements preserve
their correct relative positions, spacing, and alignment?
4. Style & Color Fidelity: Do the colors, aesthetics, and vis
ual styles match the source diagram (accounting for the anima
tion style)?
5. Proportion & Scale Fidelity: Are element sizes, aspect rat
ios, and relative scales consistent?
6. Text & Typographic Integrity: Are text labels and mathemat
ical symbols perfectly preserved without corruption, truncati
on, or hallucination?

Be extremely strict in your evaluation. Any structural mismat
ch, missing arrow, misplaced containment box, or corrupted te
xt label must lead to a noticeable drop in score.
The CRITICAL factors must induce a heavier penalty on the sco
re.

The following is NOT a penalty and MUST NOT lower any score:
  - Additional text panels, equations, step-number overlays,
titles, decorative
    annotations, or instructional labels added by the animati
on system. These are
    considered additive helpful content.

*** STYLE ADAPTER: Alpha Masking ***
In Alpha Masking, elements are revealed by fading in from tra
nsparent to opaque. After a group is revealed, previously sho
wn elements may be masked back to a lower opacity so that the
currently active group is visually emphasized. Parent contain
ers or elements with a logical/semantic reason to persist may
```

````
remain visible across steps.
You are provided with the original diagram followed by ALL frames in chronological order. Evaluate ONLY the unmasked/active regions in each frame against their corresponding region in the original diagram. Do NOT count the lower opacity or transparent masking as an element alteration or distortion.

*** OUTPUT REQUIREMENT ***
Output ONLY valid JSON matching the schema below. Reason deeply through each factor in the `assessment` dictionary first, then assign a single `visual_fidelity_score` on a scale of 0.0 to 10.0 for each frame evaluated.

```json
{
  "frame_evaluations": [
    {
      "frame_name": "string (filename of the frame evaluated)",
      "assessment": {
        "element_alteration": "Detailed reasoning regarding element presence, distortion, or style tolerance.",
        "visual_layout_fidelity": "Detailed reasoning regarding containment, flow, and structural layout.",
        "geometric_layout_fidelity": "Detailed reasoning regarding relative positions, alignment, and spacing.",
        "style_color_fidelity": "Detailed reasoning regarding colors, line weights, and visual matching.",
        "proportion_scale_fidelity": "Detailed reasoning regarding element sizes and aspect ratios.",
        "text_typographic_integrity": "Detailed reasoning regarding text legibility, corruption, or hallucination."
      },
      "visual_fidelity_score": 0.0
    }
  ]
````
````

```
}
```
```

Do not output any text before or after the JSON block.
```

Original Image



Gemini 3.1 Pro generated Video

attachment:01827fcc-b4ce-48b1-9822-4f0cc737be7f:gemini_video_alpha.mp4

```
{
  "frame_evaluations": [
    {
      "frame_name": "frame_1",
      "assessment": {
        "element_alteration": "The top-left text label 'Views: ξ_1:K' is altered to 'S: ξ_1:K', constituting a text corruption. Al
```

```
l other elements (images,
masks, depth maps, arrows,
and section labels) are pr
esent and unaltered. Style
-specific masking (opacity
changes) are tolerated and
not considered alteratio
n.",
        "visual_layout_fid
elity": "The overall visua
l layout, including contai
nment boxes, element place
ment, and directional flow
s, matches the source diag
ram exactly. The structure
of the three sections and
the internal flows are pre
served without deviatio
n.",
        "geometric_layout_
fidelity": "All elements m
aintain their correct rela
tive positions, spacing, a
nd alignment as in the sou
rce. No shifts or misalign
ments are observed in any
element.",
        "style_color_fidel
ity": "The colors, line we
ights, and visual styles
(e.g., the red arrow for L
_cross, blue arrows for re
ndering) are identical to
the source. The aesthetic
is consistent across all e
lements.",
```

```
        "proportion_scale_fidelity": "Element sizes, aspect ratios, and relative scales are consistent with the source. No resizing or distortion of elements is present.",
        "text_typographic_integrity": "The text label 'Views' is corrupted to 'S', which is a truncation and alteration of the original label. All other text labels and mathematical symbols (e.g., ξ_1:K, M_1:K, L_cross) are perfectly preserved without corruption."
      },
      "visual_fidelity_score": 8.5
    },
    {
      "frame_name": "frame_2",
      "assessment": {
        "element_alteration": "The bear statue image in the top-left section is replaced by a real bear image, which is a significant alteration of the original element. Additionally, the top-left text label 'Views: ξ_1:K' is altered to 'Gaussians: ξ_1:K'. All ot
```

```
her elements appear unalte
red but are contextually i
nconsistent due to the ima
ge change.",
        "visual_layout_fid
elity": "The overall layou
t structure (three section
s, containment boxes, and
directional flows) is main
tained. However, the conte
nt of the top-left image i
s altered, breaking the se
mantic consistency of the
diagram.",
        "geometric_layout_
fidelity": "Relative posit
ions and spacing of elemen
ts are preserved, but the
altered image content disr
upts the intended visual r
epresentation and contex
t.",
        "style_color_fidel
ity": "The diagram style
(arrows, boxes) is consist
ent, but the bear image no
w has different colors (re
al brown bear vs. white st
atue), altering the visual
style of that element.",
        "proportion_scale_
fidelity": "Element sizes
and aspect ratios are cons
istent, but the image cont
ent change affects the per
ceived scale and context o
```

```
f the bear representatio
n.",
        "text_typographic_
integrity": "The top-left
text label is corrupted fr
om 'Views' to 'Gaussians'.
All other text labels are
preserved without corrupti
on."
      },
      "visual_fidelity_sco
re": 5.0
    },
    {
      "frame_name": "frame
_3",
      "assessment": {
        "element_alteratio
n": "The active region (ri
ght panel) is expanded to
show a detailed 3D point c
loud and bear model, which
is not present in the sour
ce diagram. The source has
a compact representation o
f the refinement section,
while this frame shows an
expanded view with additio
nal elements (point cloud,
color-coded depth, et
c.).",
        "visual_layout_fid
elity": "The overall layou
t is restructured: the rig
ht panel is expanded and t
he left is minimized, devi
```

```
ating from the source's fixed three-section layout. The containment boxes and flows in the active region do not match the source.",
        "geometric_layout_fidelity": "The active region elements (point cloud, bear model) are positioned differently than in the source's compact representation. The relative alignment and spacing are inconsistent with the source.",
        "style_color_fidelity": "The active region uses a different visual style (point cloud with color gradient, detailed bear model) compared to the source's simplified 3D model and depth maps. The color scheme for depth is altered.",
        "proportion_scale_fidelity": "The active region elements are scaled up and rearranged, with different aspect ratios than in the source.",
        "text_typographic_integrity": "Text labels in the active region (e.g., 'P_1: N'', 'L_render') are present but arranged in a different layout. The sour
```

```
ce has a specific arrangem
ent of labels that is not
matched."
      },
      "visual_fidelity_sco
re": 3.0
    },
    {
      "frame_name": "frame
_4",
      "assessment": {
        "element_alteratio
n": "The active region (ri
ght panel) is expanded and
altered, similar to frame_
3. Additionally, the bear
image in the top-right of
the active region is chang
ed from a white statue to
a real brown bear, which i
s not present in the sourc
e.",
        "visual_layout_fid
elity": "Same as frame_3:
the expanded right panel d
eviates from the source's
layout. The overall struct
ure is restructured, break
ing from the source diagra
m.",
        "geometric_layout_
fidelity": "Same as frame_
3: inconsistent positionin
g and spacing in the activ
e region. Elements are rea
rranged and scaled differe
```

```
ntly.",
        "style_color_fidel
ity": "The active region h
as the same expanded style
as frame_3, plus the bear
image has different colors
(real bear vs statue), alt
ering the visual style.",
        "proportion_scale_
fidelity": "Same as frame_
3: inconsistent scaling an
d arrangement of elements
in the active region.",
        "text_typographic_
integrity": "Same as frame
_3: labels are present but
in a different layout, not
matching the source."
      },
      "visual_fidelity_sco
re": 2.5
    }
  ]
}
Average: 4.75
```

AnimateBanana generated video (TikZ)

https://drive.google.com/file/d/1-huLYZbZszOWO1k8un2MgDQFOKrUQuiv/view?usp=sharing

```
{
  "frame_evaluations": [
    {
      "frame_name": "frame
1",
      "assessment": {
        "element_alteratio
n": "All elements in the a
ctive region (left sectio
```

```
n) are present and unaltered. Input images (I_1:K, M_1:K), 3D Gaussian splatting model visualization, depth map (D_1:K), and all arrows match the original diagram. Alpha masking only affects opacity of inactive regions, which is style-tolerated.",
        "visual_layout_fidelity": "Containment box for 'Gaussians with Bear Statue' section is correctly preserved with exact directional flows (circular arrow from 3D model to input views, 'Render Depth' arrow to depth map). Structural layout matches original diagram precisely.",
        "geometric_layout_fidelity": "Relative positions of elements (input images above 3D model, 3D model above depth map) and spacing between components are identical to original. Alignment of labels and visual elements is maintained.",
        "style_color_fidelity": "Color scheme is consistent: depth map uses same thermal color gradient (red/blue), input images r
```

```
etain natural colors, mask
s remain black/white, and
blue directional arrows ma
tch original styling. Line
weights and visual aesthet
ics are preserved.",
        "proportion_scale_
fidelity": "Element sizes
and aspect ratios (e.g., d
epth map dimensions, image
grid proportions) are iden
tical to original diagram.
No scaling distortions obs
erved in active region.",
        "text_typographic_
integrity": "All text labe
ls ('Views: ξ_1:K', 'G_1:
N', 'Render Depth', 'D_1:
K', section label) and mat
hematical symbols are perf
ectly preserved without co
rruption, truncation, or h
allucination."
      },
      "visual_fidelity_sco
re": 10.0
    },
    {
      "frame_name": "frame
2",
      "assessment": {
        "element_alteratio
n": "Active region (left s
ection) contains all origi
nal elements without disto
rtion. Input views, 3D mod
```

```
el, and depth rendering co
mponents are identical to
source. Alpha masking of i
nactive regions is style-t
olerated per criteria.",
        "visual_layout_fid
elity": "Containment struc
ture and directional flows
within the left section ar
e preserved exactly. The c
ircular arrow from 3D mode
l to input views and 'Rend
er Depth' flow maintain or
iginal structural integrit
y.",
        "geometric_layout_
fidelity": "Geometric rela
tionships (e.g., spatial a
rrangement of input images
above 3D model) and spacin
g between components match
the original diagram with
pixel-perfect alignment.",
        "style_color_fidel
ity": "Visual style fideli
ty is maintained: depth ma
p color gradient, image na
tural colors, and blue arr
ow styling match the sourc
e diagram. No color shifts
or aesthetic deviations in
active region.",
        "proportion_scale_
fidelity": "Element propor
tions (e.g., relative size
of depth map to input gri
```

```
d) and aspect ratios are c
onsistent with original. N
o scaling artifacts in act
ive region.",
        "text_typographic_
integrity": "Text elements
including 'ξ_1:K', 'G_1:
N', 'D_1:K', and section l
abel 'Gaussians with Bear
Statue' are fully legible
and typographically identi
cal to source."
      },
      "visual_fidelity_sco
re": 10.0
    },
    {
      "frame_name": "frame
3",
      "assessment": {
        "element_alteratio
n": "Active region (left s
ection) shows all original
elements without alteratio
n. Depth map (D_1:K) and
'Render Depth' arrow are h
ighlighted but content mat
ches source exactly. Alpha
masking of inactive region
s is style-tolerated.",
        "visual_layout_fid
elity": "Containment box a
nd internal flow structure
are preserved. The 'Render
Depth' directional flow an
d spatial relationship bet
```

```
ween 3D model and depth map match original diagram precisely.",
        "geometric_layout_fidelity": "Relative positioning (e.g., depth map directly below 'Render Depth' arrow) and spacing are maintained without deviation. Alignment of all components matches source.",
        "style_color_fidelity": "Color fidelity is perfect: depth map uses identical thermal gradient, blue arrow styling matches source, and image colors are unaltered. No visual style deviations in active region.",
        "proportion_scale_fidelity": "Element scales (e.g., depth map dimensions relative to input images) are consistent with original. Aspect ratios preserved without distortion.",
        "text_typographic_integrity": "All text labels ('Render Depth', 'D_1:K', section label) and mathematical notation are perfectly preserved without corruption or truncation."
      },
      "visual_fidelity_sco
```

```
re": 10.0
    },
    {
      "frame_name": "frame
4",
      "assessment": {
        "element_alteratio
n": "Active region (middle
section) contains all orig
inal elements: input image
s (I_2, I_1), masks (M_2,
M_1), masked images (I_2^
B, I_1^B), depth maps (D_
2, D_1:2), and projection
components. No elements al
tered or missing.",
        "visual_layout_fid
elity": "Containment box f
or 'Inferring Depth-Guided
Inpainting Mask' section i
s correctly preserved. Two
-column layout and directi
onal flows (e.g., 'Projec
t' arrow) match original s
tructural design exactl
y.",
        "geometric_layout_
fidelity": "Relative posit
ions (e.g., input images a
bove masks, depth maps bel
ow) and spacing between el
ements are identical to so
urce. Column alignment and
component positioning are
pixel-perfect.",
        "style_color_fidel
```

```
ity": "Color scheme consistency maintained: depth maps use same thermal gradient, blue directional arrows match source, and mask visual style preserved. No aesthetic deviations in active region.",
        "proportion_scale_fidelity": "Element proportions (e.g., depth map sizes relative to input images) and aspect ratios match original diagram. No scaling distortions observed.",
        "text_typographic_integrity": "All text labels ('I_2', 'M_2', 'D_2', 'D_1:2', 'Project', section label) and mathematical symbols are perfectly preserved without corruption or truncation."
      },
      "visual_fidelity_score": 10.0
    }
  ]
}
Average: 10
```

Original Presentation

attachment:b01a33a2-f7a4-4b5d-8632-0bd827cb009f:OG_alpha.mp4


```
{
  "frame_evaluations": [
    {
      "frame_name": "slide_8",
      "assessment": {
        "element_alteration": "All active elements in the left and middle sections are present, undistorted, and match the original diagram. The faded regions (right section) are intentionally masked per animation style and not evaluated as alterations. Masks (M1:K, M2, etc.) retain correct black/white representation without distortion.",
        "visual_layout_fidelity": "Containment boxes for active sections ('Gaussians with Bear Statue' and 'Inferring Depth-Guided Inpainting Mask') maintain exact structural placement. Directional flows (arrows, projection paths) and hierarchical grouping are preserved identically to the original diagram.",
        "geometric_layout_fidelity": "Relative positions, spacing, and alignme
```

```
nt of active elements (e.
g., views I1:K/I2:K, depth
maps D1:K/D2, mask project
ions) are precisely mainta
ined. No shifts in element
positioning or flow direct
ion observed.",
        "style_color_fidel
ity": "Active elements ret
ain original color fidelit
y: depth maps (red/blue gr
adients), natural-color im
ages, and black/white mask
s. Line weights and arrow
styles (e.g., red L_render
arrow) match the source ex
actly. Faded regions are i
ntentionally masked and no
t penalized.",
        "proportion_scale_
fidelity": "Element sizes
and aspect ratios (e.g., d
epth map stacks, mask dime
nsions, image crops) are c
onsistent with the origina
l. No scaling distortions
in active regions.",
        "text_typographic_
integrity": "All text labe
ls (e.g., 'Views: ξ1:K',
'M1:K', 'G1:N', 'Render De
pth', 'D1:K') and mathemat
ical symbols are perfectly
preserved without truncati
on or corruption. Added bu
llet points are tolerable
```

```
instructional content."
      },
      "visual_fidelity_sco
re": 10.0
    },
    {
      "frame_name": "slide
_9",
      "assessment": {
        "element_alteratio
n": "All active elements i
n the middle section ('Inf
erring Depth-Guided Inpain
ting Mask') are present an
d unaltered. Faded regions
(left/right sections) are
intentionally masked per a
nimation style. Masks (M1,
M2, M1') maintain correct
black/white representation
without morphing.",
        "visual_layout_fid
elity": "Containment box f
or the active middle secti
on is structurally identic
al to the original. Direct
ional flows (e.g., 'Projec
t' arrows, mask propagatio
n paths) and hierarchical
grouping are preserved exa
ctly, including the 'Depth
-Guided Projection' pathwa
y.",
        "geometric_layout_
fidelity": "Relative posit
ions of active elements
```

```
(e.g., I2/I1 views, I2^B/I
1^B projections, D2 depth
map) maintain precise spac
ing and alignment. No geom
etric shifts observed in t
he active region.",
        "style_color_fidel
ity": "Active elements ret
ain original style: natura
l-color images, black/whit
e masks, and color-coded d
epth maps. Arrow styles
(e.g., blue 'Project' arro
w) and line weights match
the source. Faded regions
are intentionally masked a
nd not evaluated.",
        "proportion_scale_
fidelity": "Element propor
tions (e.g., mask sizes re
lative to images, depth ma
p aspect ratios) are consi
stent with the original. N
o scaling discrepancies in
active regions.",
        "text_typographic_
integrity": "All text labe
ls (e.g., 'I2', 'M2', 'I2^
B', 'D2', 'M1'') and mathe
matical symbols are intact
without corruption. Added
bullet points are tolerabl
e instructional content."
      },
      "visual_fidelity_sco
re": 10.0
```

```
    },
    {
      "frame_name": "slide
_10",
      "assessment": {
        "element_alteratio
n": "All active elements i
n the right section ('Inpa
inting-guided 3DGS Refinem
ent') are present and unal
tered. Faded regions (lef
t/middle sections) are int
entionally masked per anim
ation style. Elements like
D1^in, I1^in, and M1' reta
in correct visual properti
es without distortion.",
        "visual_layout_fid
elity": "Containment box f
or the active right sectio
n maintains exact structur
al placement. Directional
flows (e.g., 'L_cross', 'L
_render', 'Initialize' arr
ows) and hierarchical grou
ping (including cross-view
consistency loop) are pres
erved identically to the o
riginal.",
        "geometric_layout_
fidelity": "Relative posit
ions of active elements
(e.g., I2:K^P/I2:K' image
pairs, P1 Gaussian model,
D1^in depth map) maintain
precise spacing and alignm
```

```
ent. No geometric shifts i
n the active region.",
        "style_color_fidel
ity": "Active elements ret
ain original style: color-
coded depth maps, natural-
color images, and black/wh
ite masks. Critical arrow
styles (e.g., red 'L_cros
s' and 'L_render' arrows)
match the source exactly.
Faded regions are intentio
nally masked and not evalu
ated.",
        "proportion_scale_
fidelity": "Element propor
tions (e.g., depth map dim
ensions relative to image
s, Gaussian model size) ar
e consistent with the orig
inal. Aspect ratios of all
active components are pres
erved.",
        "text_typographic_
integrity": "All text labe
ls (e.g., 'I2:K^P', 'I2:
K'', 'L_cross', 'P1', 'D1^
in', 'I1^in', 'M1'') and m
athematical symbols are pe
rfectly preserved without
truncation or hallucinatio
n. Added bullet points are
tolerable instructional co
ntent."
      },
      "visual_fidelity_sco
```

```
re": 10.0
    }
  ]
}
Average Score: 10
```

Hopping Bounding Box

```
You are an expert Diagram Animation Evaluator. Your task is t
o compare the provided animation frame(s) against the origina
l source diagram to assess its visual fidelity.

You MUST evaluate the frame(s) strictly according to the foll
owing 6 criteria:

1. Element Alteration: Are all corresponding elements from th
e original diagram present, undistorted, unmorphed, and unalt
ered in the given animation frame/frames? Any style-specific
visual quirks (e.g., masking opacities, highlight boxes) MUST
be tolerated and not penalized as distortion or morphing.
2. Visual Layout Fidelity (CRITICAL): Is the overall visual l
ayout in terms of containment boxes, element placement, and d
irectional flows captured exactly as in the source diagram?
3. Geometric Layout Fidelity (CRITICAL): Do elements preserve
their correct relative positions, spacing, and alignment?
4. Style & Color Fidelity: Do the colors, aesthetics, and vis
ual styles match the source diagram (accounting for the anima
tion style)?
5. Proportion & Scale Fidelity: Are element sizes, aspect rat
ios, and relative scales consistent?
6. Text & Typographic Integrity: Are text labels and mathemat
ical symbols perfectly preserved without corruption, truncati
```

```
on, or hallucination?

Be extremely strict in your evaluation. Any structural mismat
ch, missing arrow, misplaced containment box, or corrupted te
xt label must lead to a noticeable drop in score.
The CRITICAL factors must induce a heavier penalty on the sco
re.

The following is NOT a penalty and MUST NOT lower any score:
  - Additional text panels, equations, step-number overlays,
titles, decorative
    annotations, or instructional labels added by the animati
on system. These are
    considered additive helpful content.

*** STYLE ADAPTER: Hopping Bounding Box ***
In Hopping Bounding Box, a transparent bounding box discretel
y jumps from one element or group to the next at each timeste
p, highlighting the active region.
You are provided with the original diagram followed by the LA
ST frame of the animation sequence. The highlighted bounding
box is an intentional style feature and MUST NOT be counted a
s an alteration, distortion, or morphing of the underlying di
agram elements.

*** OUTPUT REQUIREMENT ***
Output ONLY valid JSON matching the schema below. Reason deep
ly through each factor in the `assessment` dictionary first,
then assign a single `visual_fidelity_score` on a scale of 0.
0 to 10.0 for each frame evaluated.

```json
{
  "frame_evaluations": [
    {
      "frame_name": "string (filename of the frame evaluate
```
```

````
d)",
      "assessment": {
        "element_alteration": "Detailed reasoning regarding e
lement presence, distortion, or style tolerance.",
        "visual_layout_fidelity": "Detailed reasoning regardi
ng containment, flow, and structural layout.",
        "geometric_layout_fidelity": "Detailed reasoning rega
rding relative positions, alignment, and spacing.",
        "style_color_fidelity": "Detailed reasoning regarding
colors, line weights, and visual matching.",
        "proportion_scale_fidelity": "Detailed reasoning rega
rding element sizes and aspect ratios.",
        "text_typographic_integrity": "Detailed reasoning reg
arding text legibility, corruption, or hallucination."
      },
      "visual_fidelity_score": 0.0
    }
  ]
}
```

Do not output any text before or after the JSON block.
````

Original Image
````



Gemini Pro generated animation

attachment:93f6ee52-748f-431f-a43f-b0e1f24372f0:1806.mp4


```
{
  "frame_evaluations": [
    {
      "frame_name": "last_frame.png",
      "assessment": {
        "element_alteration": "Multiple critical element alterations: 'CNN Training' text is missing from Facial Feature Extractor; 'Evage lin guide' (corrupted text) and 'Rendering' steps are added to Person Dictionary; 'Feature Extraction' step is entirely removed from Person Identification; building icon in
```

```
Internet Archive is simpli
fied; vertical 'ARCHIVE' t
ext is deleted; CNN diagra
m node colors changed (blu
e instead of red). Hopping
bounding box is tolerated
as style quirk but core el
ement removals/additions v
iolate integrity.",
        "visual_layout_fid
elity": "Critical failure:
Structural layout fundamen
tally altered. Original 2x
2 containment boxes retain
positions but internal flo
ws are rerouted. Red arrow
from CNN output now connec
ts to 'Face Verification'
instead of 'Feature Extrac
tion'. 'Filtering' step in
Person Dictionary incorrec
tly feeds into 'Rendering'
rather than Person Identif
ication. Person Identifica
tion workflow is broken by
missing 'Feature Extractio
n' step, changing the enti
re directional flow (Face
Detection → Face Verificat
ion → Visualization).",
        "geometric_layout_
fidelity": "Critical failu
re: Relative positions and
alignments severely distor
ted. Added 'Evage lin guid
e' and 'Rendering' steps i
```

```
n Person Dictionary shift
all internal elements vert
ically. Removal of 'Featur
e Extraction' in Person Id
entification causes adjace
nt elements to collapse an
d reposition. Arrow endpoi
nts misaligned (e.g., red
arrow from CNN now termina
tes at wrong step). Stack
of images in Person Dictio
nary introduced at incorre
ct geometric position.",
        "style_color_fidel
ity": "Significant style d
eviation: Facial Feature E
xtractor CNN nodes use blu
e instead of original red;
Internet Archive building
icon loses architectural d
etail; web content browser
redesign alters visual sty
le; Person Dictionary back
ground shade changed. Line
weights inconsistent (e.
g., thicker arrows in anim
ation). Color scheme for s
ection backgrounds is appr
oximately maintained but i
nternal aesthetics mismatc
hed.",
        "proportion_scale_
fidelity": "Severe scaling
issues: CNN diagram enlarg
ed disproportionately in F
acial Feature Extractor.
```

```
'Rendering' box in Person Dictionary is oversized relative to other steps. Web content browser in Internet Archive scaled differently. Person Identification steps compressed due to missing 'Feature Extraction' step, altering aspect ratios of all elements in that section.",
        "text_typographic_integrity": "Critical text corruption: 'CNN Training' deleted; 'Evage lin guide' (likely 'Evasion guide' typo) added; 'Key 2' introduced; vertical 'ARCHIVE' text removed; 'Feature Extraction' label missing. Font styles inconsistent (e.g., 'Rendering' uses different weight). All text changes constitute hallucinations or truncations violating original labeling."
      },
      "visual_fidelity_score": 1.5
    }
  ]
}
```

Sliding Bounding Box

```
You are an expert Diagram Animation Evaluator. Your task is t
o compare the provided animation frame(s) against the origina
l source diagram to assess its visual fidelity.

You MUST evaluate the frame(s) strictly according to the foll
owing 6 criteria:

1. Element Alteration: Are all corresponding elements from th
e original diagram present, undistorted, unmorphed, and unalt
ered in the given animation frame/frames? Any style-specific
visual quirks (e.g., masking opacities, highlight boxes) MUST
be tolerated and not penalized as distortion or morphing.
2. Visual Layout Fidelity (CRITICAL): Is the overall visual l
ayout in terms of containment boxes, element placement, and d
irectional flows captured exactly as in the source diagram?
3. Geometric Layout Fidelity (CRITICAL): Do elements preserve
their correct relative positions, spacing, and alignment?
4. Style & Color Fidelity: Do the colors, aesthetics, and vis
ual styles match the source diagram (accounting for the anima
tion style)?
5. Proportion & Scale Fidelity: Are element sizes, aspect rat
ios, and relative scales consistent?
6. Text & Typographic Integrity: Are text labels and mathemat
ical symbols perfectly preserved without corruption, truncati
on, or hallucination?

Be extremely strict in your evaluation. Any structural mismat
ch, missing arrow, misplaced containment box, or corrupted te
xt label must lead to a noticeable drop in score.
The CRITICAL factors must induce a heavier penalty on the sco
re.

The following is NOT a penalty and MUST NOT lower any score:
  - Additional text panels, equations, step-number overlays,
```

```
titles, decorative
    annotations, or instructional labels added by the animati
on system. These are
    considered additive helpful content.

*** STYLE ADAPTER: Sliding Bounding Box ***
In Sliding Bounding Box, a transparent bounding box smoothly
slides across the canvas to reach the currently active elemen
t or group.
You are provided with the original diagram followed by the LA
ST frame of the animation sequence. The highlighted bounding
box is an intentional style feature and MUST NOT be counted a
s an alteration, distortion, or morphing of the underlying di
agram elements.

*** OUTPUT REQUIREMENT ***
Output ONLY valid JSON matching the schema below. Reason deep
ly through each factor in the `assessment` dictionary first,
then assign a single `visual_fidelity_score` on a scale of 0.
0 to 10.0 for each frame evaluated.

```json
{
  "frame_evaluations": [
    {
      "frame_name": "string (filename of the frame evaluate
d)",
      "assessment": {
        "element_alteration": "Detailed reasoning regarding e
lement presence, distortion, or style tolerance.",
        "visual_layout_fidelity": "Detailed reasoning regardi
ng containment, flow, and structural layout.",
        "geometric_layout_fidelity": "Detailed reasoning rega
rding relative positions, alignment, and spacing.",
        "style_color_fidelity": "Detailed reasoning regarding
colors, line weights, and visual matching.",
```
```

```
        "proportion_scale_fidelity": "Detailed reasoning rega
rding element sizes and aspect ratios.",
        "text_typographic_integrity": "Detailed reasoning reg
arding text legibility, corruption, or hallucination."
      },
      "visual_fidelity_score": 0.0
    }
  ]
}
```

Do not output any text before or after the JSON block.
```

## Colour Pop

```
You are an expert Diagram Animation Evaluator. Your task is t
o compare the provided animation frame(s) against the origina
l source diagram to assess its visual fidelity.

You MUST evaluate the frame(s) strictly according to the foll
owing 6 criteria:

1. Element Alteration: Are all corresponding elements from th
e original diagram present, undistorted, unmorphed, and unalt
ered in the given animation frame/frames? Any style-specific
visual quirks (e.g., masking opacities, highlight boxes) MUST
be tolerated and not penalized as distortion or morphing.
2. Visual Layout Fidelity (CRITICAL): Is the overall visual l
ayout in terms of containment boxes, element placement, and d
irectional flows captured exactly as in the source diagram?
3. Geometric Layout Fidelity (CRITICAL): Do elements preserve
their correct relative positions, spacing, and alignment?
```
```

```
4. Style & Color Fidelity: Do the colors, aesthetics, and visual styles match the source diagram (accounting for the animation style)?
5. Proportion & Scale Fidelity: Are element sizes, aspect ratios, and relative scales consistent?
6. Text & Typographic Integrity: Are text labels and mathematical symbols perfectly preserved without corruption, truncation, or hallucination?

Be extremely strict in your evaluation. Any structural mismatch, missing arrow, misplaced containment box, or corrupted text label must lead to a noticeable drop in score.
The CRITICAL factors must induce a heavier penalty on the score.

The following is NOT a penalty and MUST NOT lower any score:
  - Additional text panels, equations, step-number overlays, titles, decorative
    annotations, or instructional labels added by the animation system. These are
    considered additive helpful content.

*** STYLE ADAPTER: Colour Pop ***
In Colour Pop, the entire diagram starts in greyscale. At each step, the currently active group of elements transitions from greyscale to full colour while the rest remains grey.
You are provided with the original diagram followed by the LAST frame of the animation sequence (which represents the fully colored state). Evaluate whether all colors, elements, and layout match the original diagram exactly.

*** OUTPUT REQUIREMENT ***
Output ONLY valid JSON matching the schema below. Reason deeply through each factor in the `assessment` dictionary first, then assign a single `visual_fidelity_score` on a scale of 0.0 to 10.0 for each frame evaluated.
```

```json
{
  "frame_evaluations": [
    {
      "frame_name": "string (filename of the frame evaluated)",
      "assessment": {
        "element_alteration": "Detailed reasoning regarding element presence, distortion, or style tolerance.",
        "visual_layout_fidelity": "Detailed reasoning regarding containment, flow, and structural layout.",
        "geometric_layout_fidelity": "Detailed reasoning regarding relative positions, alignment, and spacing.",
        "style_color_fidelity": "Detailed reasoning regarding colors, line weights, and visual matching.",
        "proportion_scale_fidelity": "Detailed reasoning regarding element sizes and aspect ratios.",
        "text_typographic_integrity": "Detailed reasoning regarding text legibility, corruption, or hallucination."
      },
      "visual_fidelity_score": 0.0
    }
  ]
}
```

Do not output any text before or after the JSON block.
```

# Judging animation video

Here we are using the animation video as a whole and feeding it to the Judge model to comment on the same criteria but now accounting for the smoothness of
```

the video

Judge: Gemini 3.5 Flash

Have general rules and attune to style-specific adapters

Implementation Idea: The generic instructions at the start of the prompt and before and after the style adapter are common in every prompt. The text of the style adapter is added based on the chosen style for the animation. So the gating must be done accurately based on the style

The Prompts are as follows:

Progressive Reveal

```
You are an expert Diagram Animation Evaluator. Your task is t
o compare a video of an animated diagram sequence against the
original source diagram to assess its visual fidelity.

You are shown the original source diagram as a still image, f
ollowed by a video of the complete animation sequence. Watch
the full video, not just its final state — a defect that appe
ars and is later corrected is still evidence of an unfaithful
animation and must be reflected in your assessment, even if t
he video ends on a clean frame.

You MUST evaluate the video strictly according to the followi
ng 6 criteria:

1. Element Alteration: Are all corresponding elements from th
e original diagram present, undistorted, unmorphed, and unalt
ered at every point in the video where they are visible? Any
style-specific visual quirks (e.g., masking opacities, highli
ght boxes) MUST be tolerated and not penalized as distortion
```

```
or morphing.
2. Visual Layout Fidelity (CRITICAL): Is the overall visual l
ayout in terms of containment boxes, element placement, and d
irectional flows captured exactly as in the source diagram, t
hroughout the video?
3. Geometric Layout Fidelity (CRITICAL): Do elements preserve
their correct relative positions, spacing, and alignment thro
ughout the video?
4. Style & Color Fidelity: Do the colors, aesthetics, and vis
ual styles match the source diagram (accounting for the anima
tion style)?
5. Proportion & Scale Fidelity: Are element sizes, aspect rat
ios, and relative scales consistent throughout the video?
6. Text & Typographic Integrity: Are text labels and mathemat
ical symbols perfectly preserved without corruption, truncati
on, or hallucination at any point in the video?

Be extremely strict in your evaluation. Any structural mismat
ch, missing arrow, misplaced containment box, or corrupted te
xt label — even one that appears only briefly mid-animation —
must lead to a noticeable drop in score.
The CRITICAL factors must induce a heavier penalty on the sco
re.

The following is NOT a penalty and MUST NOT lower any score:
  - Additional text panels, equations, step-number overlays,
titles, decorative
    annotations, or instructional labels added by the animati
on system. These are
    considered additive helpful content.

*** STYLE ADAPTER: Progressive Reveal ***
In Progressive Reveal, elements of the diagram are cumulative
ly added to the canvas one group at a time. Once revealed, el
ements remain visible for the rest of the animation. Nothing
is hidden or masked after being shown.
```

````
Evaluate whether the diagram, as it accumulates over the cour
se of the video, is faithful to the source at every stage — a
nd whether the fully revealed end state is completely faithfu
l to the original source diagram.

*** OUTPUT REQUIREMENT ***
Output ONLY valid JSON matching the schema below. Reason deep
ly through each factor in the `assessment` dictionary first,
then assign a single `visual_fidelity_score` on a scale of 0.
0 to 10.0 for the video as a whole.

```json
{
  "video_evaluation": {
    "assessment": {
      "element_alteration": "Detailed reasoning regarding ele
ment presence, distortion, or style tolerance.",
      "visual_layout_fidelity": "Detailed reasoning regarding
containment, flow, and structural layout.",
      "geometric_layout_fidelity": "Detailed reasoning regard
ing relative positions, alignment, and spacing.",
      "style_color_fidelity": "Detailed reasoning regarding c
olors, line weights, and visual matching.",
      "proportion_scale_fidelity": "Detailed reasoning regard
ing element sizes and aspect ratios.",
      "text_typographic_integrity": "Detailed reasoning regar
ding text legibility, corruption, or hallucination."
    },
    "temporal_defects_observed": "List and briefly describe a
ny defect visible only at some point mid-video (approximate t
imestamp if possible), even if resolved by the end. Write 'No
ne observed' if there were none.",
    "visual_fidelity_score": 0.0
  }
}
```
````

```
Do not output any text before or after the JSON block.
```

Gemini Pro animation video

attachment:3469c196-bd65-45bf-8914-3ad5e4c24a8a:CVPR_2025_pipe00010.mp4


```
{
  "frame_evaluations": [
    {
      "frame_name": "animated_frame.png",
      "assessment": {
        "element_alteration": "Several critical elements are altered or completely missing. The entire 'Distortion' block (JPEG compression, crop, rotation) is absent. The watermark key icon on the 3D model is changed to a blue shield lock, and the Gaussian shapes have been stylized differently.",
        "visual_layout_fidelity": "The visual layout is severely modified. The original two-branch pipeline under 'Unauthorized Use' is collapsed into a single, logically incorrect path. The 'Identification' step is moved from the bottom blue container to the middle grey container.",
        "geometric_layout_fidelity": "Relative posit
```

```
ioning and alignment are h
ighly compromised. Multipl
e directional arrows have
been redirected or remove
d, disrupting the geometri
c layout and flow of the o
riginal diagram.",
        "style_color_fidel
ity": "The core container
colors are maintained, but
individual icon designs, i
mage contents (fountain re
placing garden pictures in
the distortion box), and g
raphics show noticeable st
yling deviations.",
        "proportion_scale_
fidelity": "Proportions ar
e highly inconsistent due
to the omission of major c
omponents, leaving the rem
aining boxes and elements
awkwardly sized and spaced
compared to the source.",
        "text_typographic_
integrity": "Several impor
tant textual annotations,
including 'Distortion', 'J
PEG Compression', 'Crop',
'Rotation', and 'Distorted
3D-GS', are entirely missi
ng."
      },
      "visual_fidelity_sco
re": 3.5
    }
```

```
    ]
}
```

Alpha Masking

```
You are an expert Diagram Animation Evaluator. Your task is t
o compare a video of an animated diagram sequence against the
original source diagram to assess its visual fidelity.

You are shown the original source diagram as a still image, f
ollowed by a video of the complete animation sequence. Watch
the full video, not just its final state — a defect that appe
ars and is later corrected is still evidence of an unfaithful
animation and must be reflected in your assessment, even if t
he video ends on a clean frame.

You MUST evaluate the video strictly according to the followi
ng 6 criteria:

1. Element Alteration: Are all corresponding elements from th
e original diagram present, undistorted, unmorphed, and unalt
ered at every point in the video where they are visible? Any
style-specific visual quirks (e.g., masking opacities, highli
ght boxes) MUST be tolerated and not penalized as distortion
or morphing.
2. Visual Layout Fidelity (CRITICAL): Is the overall visual l
ayout in terms of containment boxes, element placement, and d
irectional flows captured exactly as in the source diagram, t
hroughout the video?
3. Geometric Layout Fidelity (CRITICAL): Do elements preserve
their correct relative positions, spacing, and alignment thro
ughout the video?
```

```
4. Style & Color Fidelity: Do the colors, aesthetics, and vis
ual styles match the source diagram (accounting for the anima
tion style)?
5. Proportion & Scale Fidelity: Are element sizes, aspect rat
ios, and relative scales consistent throughout the video?
6. Text & Typographic Integrity: Are text labels and mathemat
ical symbols perfectly preserved without corruption, truncati
on, or hallucination at any point in the video?

Be extremely strict in your evaluation. Any structural mismat
ch, missing arrow, misplaced containment box, or corrupted te
xt label — even one that appears only briefly mid-animation —
must lead to a noticeable drop in score.
The CRITICAL factors must induce a heavier penalty on the sco
re.

The following is NOT a penalty and MUST NOT lower any score:
  - Additional text panels, equations, step-number overlays,
titles, decorative
    annotations, or instructional labels added by the animati
on system. These are
    considered additive helpful content.

*** STYLE ADAPTER: Alpha Masking ***
In Alpha Masking, elements are revealed by fading in from tra
nsparent to opaque. After a group is revealed, previously sho
wn elements may be masked back to a lower opacity so that the
currently active group is visually emphasized. Parent contain
ers or elements with a logical/semantic reason to persist may
remain visible across steps.
At every point in the video, evaluate ONLY the unmasked/curre
ntly emphasized regions against their corresponding region in
the original diagram. Do NOT count the lower opacity or trans
parent masking of previously-shown content as an element alte
ration or distortion. However, DO flag any genuine morphing,
hallucination, or corruption that appears in a currently-emph
```

```
asized region at any point during the fade cycle, even if tha
t region is later masked back down and the issue becomes invi
sible by the video's end — a defect hidden by a later mask is
still a defect.

*** OUTPUT REQUIREMENT ***
Output ONLY valid JSON matching the schema below. Reason deep
ly through each factor in the `assessment` dictionary first,
then assign a single `visual_fidelity_score` on a scale of 0.
0 to 10.0 for the video as a whole.

```json
{
  "video_evaluation": {
    "assessment": {
      "element_alteration": "Detailed reasoning regarding ele
ment presence, distortion, or style tolerance.",
      "visual_layout_fidelity": "Detailed reasoning regarding
containment, flow, and structural layout.",
      "geometric_layout_fidelity": "Detailed reasoning regard
ing relative positions, alignment, and spacing.",
      "style_color_fidelity": "Detailed reasoning regarding c
olors, line weights, and visual matching.",
      "proportion_scale_fidelity": "Detailed reasoning regard
ing element sizes and aspect ratios.",
      "text_typographic_integrity": "Detailed reasoning regar
ding text legibility, corruption, or hallucination."
    },
    "temporal_defects_observed": "List and briefly describe a
ny defect visible only at some point mid-video (approximate t
imestamp if possible), even if resolved by the end. Write 'No
ne observed' if there were none.",
    "visual_fidelity_score": 0.0
  }
}
```
```

```
Do not output any text before or after the JSON block.
```

Gemini 3.1 Pro animation video

attachment:4fb80981-bd32-4b34-9e87-f9fa4cfaaa9d:gemini_video_alpha.mp4


```
{
  "video_evaluation": {
    "assessment": {
      "element_alteration": "Major alterations occur throughout the video. The stone bear statue is replaced with a real grizzly bear, and a white polar bear is introduced in the latter stages. The point cloud P1 is transformed from a grey surface to a rainbow point cloud.",
      "visual_layout_fidelity": "The entire structural layout of the third section ('Inpainting-guided 3DGS Refinement') is altered, introducing new branches, different images, and a highly modified flow compared to the source diagram.",
      "geometric_layout_fidelity": "Elements are rearranged and repositioned to fit entir
```

```
ely new pipelines, com
pletely disrupting the
alignment and spatial
relationships of the o
riginal design.",
      "style_color_fid
elity": "The visual st
yle is inconsistent; n
atural tones and grays
cale elements are repl
aced with bright multi
-colored visualization
s, and the photographi
c content of the image
s is fundamentally cha
nged.",
      "proportion_scal
e_fidelity": "The scal
e and sizes of various
containment boxes and
image panels are alter
ed significantly to ac
commodate the newly in
troduced components an
d branches.",
      "text_typographi
c_integrity": "Labels
and mathematical notat
ions are modified or c
ompletely hallucinated
(e.g., introducing lab
els like I_11, I'_{22:
K}$, $G'_{1:N'}$, et
c.) compared to the or
iginal text."
    },
```

```
    "temporal_defects_observed": "The visual and structural content undergoes continuous, unfaithful transformation starting around 00:01 and completely changes layout by 00:06, departing from the reference diagram entirely.",
    "visual_fidelity_score": 3.0
  }
}
```

AnimateBanana Animation Video (TikZ)

https://drive.google.com/file/d/1-huLYZbZszOWO1k8un2MgDQFOKrUQuiv/view?usp=sharing

```
{
  "video_evaluation": {
    "assessment": {
      "element_alteration": "The elements in the active regions appear clean and undistorted. Visual features are accurately retained throughout the step-by-step presentation.",
      "visual_layout_fidelity": "The layout structure, process flow, and grouping zones are represented accurately in accordance with the source diagram.",
      "geometric_layout_fidelity": "Relative positio
```

```
ning, alignments, and spat
ial relationships between
diagram components are pre
served without shifting.",
      "style_color_fidelit
y": "Color accuracy, strok
e styles, and general desi
gn aesthetics match the or
iginal diagram.",
      "proportion_scale_fi
delity": "Aspect ratios, c
omponent sizes, and scales
remain consistent across t
he sequence.",
      "text_typographic_in
tegrity": "Labels, subscri
pts, and mathematical symb
ols are legible and free o
f corruption or rendering
errors."
    },
    "temporal_defects_obse
rved": "None observed",
    "visual_fidelity_scor
e": 7.6
  }
}
```

Colour Pop

```
You are an expert Diagram Animation Evaluator. Your task is t
o compare a video of an animated diagram sequence against the
```

```
original source diagram to assess its visual fidelity.

You are shown the original source diagram as a still image, f
ollowed by a video of the complete animation sequence. Watch
the full video, not just its final state — a defect that appe
ars and is later corrected is still evidence of an unfaithful
animation and must be reflected in your assessment, even if t
he video ends on a clean frame.

You MUST evaluate the video strictly according to the followi
ng 6 criteria:

1. Element Alteration: Are all corresponding elements from th
e original diagram present, undistorted, unmorphed, and unalt
ered at every point in the video where they are visible? Any
style-specific visual quirks (e.g., masking opacities, highli
ght boxes) MUST be tolerated and not penalized as distortion
or morphing.
2. Visual Layout Fidelity (CRITICAL): Is the overall visual l
ayout in terms of containment boxes, element placement, and d
irectional flows captured exactly as in the source diagram, t
hroughout the video?
3. Geometric Layout Fidelity (CRITICAL): Do elements preserve
their correct relative positions, spacing, and alignment thro
ughout the video?
4. Style & Color Fidelity: Do the colors, aesthetics, and vis
ual styles match the source diagram (accounting for the anima
tion style)?
5. Proportion & Scale Fidelity: Are element sizes, aspect rat
ios, and relative scales consistent throughout the video?
6. Text & Typographic Integrity: Are text labels and mathemat
ical symbols perfectly preserved without corruption, truncati
on, or hallucination at any point in the video?

Be extremely strict in your evaluation. Any structural mismat
ch, missing arrow, misplaced containment box, or corrupted te
```

```
xt label — even one that appears only briefly mid-animation —
must lead to a noticeable drop in score.
The CRITICAL factors must induce a heavier penalty on the sco
re.

The following is NOT a penalty and MUST NOT lower any score:
  - Additional text panels, equations, step-number overlays,
titles, decorative
    annotations, or instructional labels added by the animati
on system. These are
    considered additive helpful content.

*** STYLE ADAPTER: Colour Pop ***
In Colour Pop, the entire diagram starts in greyscale. At eac
h step, the currently active group of elements transitions fr
om greyscale to full colour while the rest remains grey. Do N
OT penalize Style & Color Fidelity for any region that is sti
ll greyscale because it has not yet been reached by the anima
tion.
Evaluate whether the diagram, as elements transition to full
color over the course of the video, remains faithful to the s
ource at every stage — and whether the fully colored end stat
e matches the original diagram exactly in color, elements, an
d layout.

*** OUTPUT REQUIREMENT ***
Output ONLY valid JSON matching the schema below. Reason deep
ly through each factor in the `assessment` dictionary first,
then assign a single `visual_fidelity_score` on a scale of 0.
0 to 10.0 for the video as a whole.

```json
{
  "video_evaluation": {
    "assessment": {
      "element_alteration": "Detailed reasoning regarding ele
```
```

```
ment presence, distortion, or style tolerance.",
      "visual_layout_fidelity": "Detailed reasoning regarding
containment, flow, and structural layout.",
      "geometric_layout_fidelity": "Detailed reasoning regard
ing relative positions, alignment, and spacing.",
      "style_color_fidelity": "Detailed reasoning regarding c
olors, line weights, and visual matching.",
      "proportion_scale_fidelity": "Detailed reasoning regard
ing element sizes and aspect ratios.",
      "text_typographic_integrity": "Detailed reasoning regar
ding text legibility, corruption, or hallucination."
    },
    "temporal_defects_observed": "List and briefly describe a
ny defect visible only at some point mid-video (approximate t
imestamp if possible), even if resolved by the end. Write 'No
ne observed' if there were none.",
    "visual_fidelity_score": 0.0
  }
}
` ` `

Do not output any text before or after the JSON block.
```

Hopping Bounding Box

```
You are an expert Diagram Animation Evaluator. Your task is t
o compare a video of an animated diagram sequence against the
original source diagram to assess its visual fidelity.

You are shown the original source diagram as a still image, f
ollowed by a video of the complete animation sequence. Watch
the full video, not just its final state — a defect that appe
```

```
ars and is later corrected is still evidence of an unfaithful
animation and must be reflected in your assessment, even if t
he video ends on a clean frame.

You MUST evaluate the video strictly according to the followi
ng 6 criteria:

1. Element Alteration: Are all corresponding elements from th
e original diagram present, undistorted, unmorphed, and unalt
ered at every point in the video where they are visible? Any
style-specific visual quirks (e.g., masking opacities, highli
ght boxes) MUST be tolerated and not penalized as distortion
or morphing.
2. Visual Layout Fidelity (CRITICAL): Is the overall visual l
ayout in terms of containment boxes, element placement, and d
irectional flows captured exactly as in the source diagram, t
hroughout the video?
3. Geometric Layout Fidelity (CRITICAL): Do elements preserve
their correct relative positions, spacing, and alignment thro
ughout the video?
4. Style & Color Fidelity: Do the colors, aesthetics, and vis
ual styles match the source diagram (accounting for the anima
tion style)?
5. Proportion & Scale Fidelity: Are element sizes, aspect rat
ios, and relative scales consistent throughout the video?
6. Text & Typographic Integrity: Are text labels and mathemat
ical symbols perfectly preserved without corruption, truncati
on, or hallucination at any point in the video?

Be extremely strict in your evaluation. Any structural mismat
ch, missing arrow, misplaced containment box, or corrupted te
xt label — even one that appears only briefly mid-animation —
must lead to a noticeable drop in score.
The CRITICAL factors must induce a heavier penalty on the sco
re.
```

```
The following is NOT a penalty and MUST NOT lower any score:
  - Additional text panels, equations, step-number overlays, titles, decorative
    annotations, or instructional labels added by the animation system. These are
    considered additive helpful content.

*** STYLE ADAPTER: Hopping Bounding Box ***
In Hopping Bounding Box, a transparent bounding box discretely jumps from one element or group to the next at each timestep, highlighting the active region. The bounding box itself is an intentional style feature and MUST NOT be counted as an alteration, distortion, or morphing of the underlying diagram elements.
Evaluate whether the diagram elements underneath the bounding box remain faithful to the source throughout the video, and whether the final state of the diagram matches the original exactly.

*** OUTPUT REQUIREMENT ***
Output ONLY valid JSON matching the schema below. Reason deeply through each factor in the `assessment` dictionary first, then assign a single `visual_fidelity_score` on a scale of 0.0 to 10.0 for the video as a whole.

```json
{
  "video_evaluation": {
    "assessment": {
      "element_alteration": "Detailed reasoning regarding element presence, distortion, or style tolerance.",
      "visual_layout_fidelity": "Detailed reasoning regarding containment, flow, and structural layout.",
      "geometric_layout_fidelity": "Detailed reasoning regarding relative positions, alignment, and spacing.",
      "style_color_fidelity": "Detailed reasoning regarding c
```
```

```
olors, line weights, and visual matching.",
      "proportion_scale_fidelity": "Detailed reasoning regard
ing element sizes and aspect ratios.",
      "text_typographic_integrity": "Detailed reasoning regar
ding text legibility, corruption, or hallucination."
    },
    "temporal_defects_observed": "List and briefly describe a
ny defect visible only at some point mid-video (approximate t
imestamp if possible), even if resolved by the end. Write 'No
ne observed' if there were none.",
    "visual_fidelity_score": 0.0
  }
}
```

Do not output any text before or after the JSON block.
```

## Sliding Bounding Box

```
You are an expert Diagram Animation Evaluator. Your task is t
o compare a video of an animated diagram sequence against the
original source diagram to assess its visual fidelity.

You are shown the original source diagram as a still image, f
ollowed by a video of the complete animation sequence. Watch
the full video, not just its final state — a defect that appe
ars and is later corrected is still evidence of an unfaithful
animation and must be reflected in your assessment, even if t
he video ends on a clean frame.

You MUST evaluate the video strictly according to the followi
ng 6 criteria:

1. Element Alteration: Are all corresponding elements from th
```
```

```
e original diagram present, undistorted, unmorphed, and unalt
ered at every point in the video where they are visible? Any
style-specific visual quirks (e.g., masking opacities, highli
ght boxes) MUST be tolerated and not penalized as distortion
or morphing.
2. Visual Layout Fidelity (CRITICAL): Is the overall visual l
ayout in terms of containment boxes, element placement, and d
irectional flows captured exactly as in the source diagram, t
hroughout the video?
3. Geometric Layout Fidelity (CRITICAL): Do elements preserve
their correct relative positions, spacing, and alignment thro
ughout the video?
4. Style & Color Fidelity: Do the colors, aesthetics, and vis
ual styles match the source diagram (accounting for the anima
tion style)?
5. Proportion & Scale Fidelity: Are element sizes, aspect rat
ios, and relative scales consistent throughout the video?
6. Text & Typographic Integrity: Are text labels and mathemat
ical symbols perfectly preserved without corruption, truncati
on, or hallucination at any point in the video?

Be extremely strict in your evaluation. Any structural mismat
ch, missing arrow, misplaced containment box, or corrupted te
xt label — even one that appears only briefly mid-animation —
must lead to a noticeable drop in score.
The CRITICAL factors must induce a heavier penalty on the sco
re.

The following is NOT a penalty and MUST NOT lower any score:
  - Additional text panels, equations, step-number overlays,
titles, decorative
    annotations, or instructional labels added by the animati
on system. These are
    considered additive helpful content.

*** STYLE ADAPTER: Sliding Bounding Box ***
```

````
In Sliding Bounding Box, a transparent bounding box smoothly slides across the canvas to reach the currently active element or group. The bounding box itself is an intentional style feature and MUST NOT be counted as an alteration, distortion, or morphing of the underlying diagram elements.
Evaluate whether the diagram elements underneath and around the sliding box remain faithful to the source throughout the video, and whether the final state of the diagram matches the original exactly.

*** OUTPUT REQUIREMENT ***
Output ONLY valid JSON matching the schema below. Reason deeply through each factor in the `assessment` dictionary first, then assign a single `visual_fidelity_score` on a scale of 0.0 to 10.0 for the video as a whole.

```json
{
  "video_evaluation": {
    "assessment": {
      "element_alteration": "Detailed reasoning regarding element presence, distortion, or style tolerance.",
      "visual_layout_fidelity": "Detailed reasoning regarding containment, flow, and structural layout.",
      "geometric_layout_fidelity": "Detailed reasoning regarding relative positions, alignment, and spacing.",
      "style_color_fidelity": "Detailed reasoning regarding colors, line weights, and visual matching.",
      "proportion_scale_fidelity": "Detailed reasoning regarding element sizes and aspect ratios.",
      "text_typographic_integrity": "Detailed reasoning regarding text legibility, corruption, or hallucination."
    },
    "temporal_defects_observed": "List and briefly describe any defect visible only at some point mid-video (approximate timestamp if possible), even if resolved by the end. Write 'No
````
````

```
ne observed' if there were none.",
    "visual_fidelity_score": 0.0
  }
}
```

Do not output any text before or after the JSON block.
```
```